

Yatin Dandi

THIRD YEAR UNDERGRADUATE · COMPUTER SCIENCE AND ENGINEERING

IIT Kanpur

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Education

Indian Institute of Technology Kanpur

BACHELOR OF TECHNOLOGY, MAJOR IN COMPUTER SCIENCE AND ENGINEERING

Kanpur, India

July, 2017 - Present

- **Cumulative Performance Index : 9.8/10.0**
- **Minors: Statistical Mechanics**

Interests

Deep Generative Models, Meta Learning, Probabilistic Machine Learning.

Publications

Jointly Trained Image and Video Generation using Residual Vectors

Colorado, USA

YATIN DANDI, ANIKET DAS, SOUMYE SINGHAL, VINAY P. NAMBOODIRI, PIYUSH RAI

March, 2020

2020 Winter Conference on Applications of Computer Vision (WACV '20)

Honors & Awards

- 2017 **Aditya Birla Scholarship**, Awarded to 15 students from all IITs
- 2017 **Academic Excellence Award**, Awarded for exceptional performance
- 2016 **KVPY Scholarship, 2016**, Indian Institute of Science
- 2015 **NTSE Scholarship**, Government of India
- 2017 **All India Rank 135**, JEE Advanced 2017
- 2016 **Selected for Indian National Physics Olympiad**, HBCSE
- 2016 **Selected for Indian National Chemistry Olympiad**, HBCSE

Mumbai, India

IIT Kanpur

Bangalore

India

India

Mumbai, India

Mumbai, India

Experience

Adversarial Joint Distribution Matching with Reconstructive Feedback

IIT Kanpur

UNDERGRADUATE PROJECT UNDER PROF. PIYUSH RAI

August 2019 - Present

- Proposed a novel approach to introduce reconstructive feedback in the ALIGAN/BIGAN framework by modifying the discriminator's objective to correctly identify more than just two joint distributions of image-latent vector pairs
- Implemented baseline and proposed models in Pytorch.
- Performed quantitative and qualitative evaluations over baselines on SVHN and CelebA datasets.
- Results indicate significant improvements over baseline in reconstruction quality and representation learning.

Joint Image and Video Generation using Residual Vectors

IIT Kanpur

UNDERGRADUATE PROJECT UNDER PROF. PIYUSH RAI AND PROF. VINAY NAMBOODIRI

May 2019 - July 2019

- Designed a novel modelling technique for joint image and video generation that simultaneously learns to map latent variables with a fixed prior onto real images and interpolate over images to generate videos.
- Implemented Variational Autoencoder and Generative Adversarial Networks based video generation models in Pytorch utilizing the proposed technique.
- Performed quantitative and qualitative evaluations over previous VAE and GAN based video generation approaches.
- Accepted in WACV 2020.

Disentangled Representation Learning using Generative Models

IIT Kanpur

COURSE PROJECT UNDER PROF. PIYUSH RAI

December 2018 - May 2019

- Studied various approaches for learning disentangled representations of sequential data such as using new adversarial loss terms, factorized hierarchical priors and exploiting the probabilistic model and architecture of the LSTM based autoencoder to promote disentanglement.
- Implemented a Variational Autoencoder model for disentangling of time invariant content and dynamics in sequential data (Mandt et al.) using Pytorch and experimented with modifications in the probabilistic model.
- Modified existing models for video generation such as MocoGAN, SVG with learned prior to improve generation and disentanglement

New York Office, IIT Kanpur

IIT Kanpur

MACHINE LEARNING FOR LARGE SCALE LOGISTICS PLATFORM, UNDER PROF. MANINDRA AGARWAL

May 2018 - July 2018

- Implemented a state of the art algorithm for online collaborative filtering based on Fast Matrix Factorization for Online Recommendation with Implicit Feedback (He et al.) using Numpy and improved the model with sentiment and frequency dependent weighting schemes.
- Used Kafka for real-time data processing and simulated interactions using locust.
- Implemented a recommender system based on deep autoencoders and compared the results with other models using metrics such as hit ratio.
- Implemented a Bidirectional LSTM model using Keras for sentiment analysis of user comments.
- Trained the Latent Dirichlet allocation model on Wikipedia articles for automatic extraction of topics.

Social Situation Inference in Simple Animated Shapes

IIT Kanpur

UNDERGRADUATE PROJECT UNDER PROF. NISHEETH SRIVASTAVA

May 2018 - July 2018

- The aim was to characterise the nature and granularity of information about social interactions accessible to human observers from simple visual displays.
- Developed an animation engine capable of producing a vast range of social situations involving autonomous agents having long term and short term goals.
- The physical parameters were inspired by the valence-arousal model and were designed to represent independent factors, together covering the entire emotion space.
- The motion of autonomous characters was based on work by Reynolds, C. W. and the social force model for crowd behavior simulation.
- Performed experiments with human subjects to determine how subjective attribution of emotions to specific situations varies as a function of the generative models parameters.

Deep Reinforcement Learning for Atari Games

IIT Kanpur

ASSOCIATION FOR COMPUTING ACTIVITIES, IIT KANPUR

February 2018 - May 2018

- Used Numpy to implement various reinforcement learning algorithms such as Dynamic Programming (Policy and Value iteration), Monte Carlo (Epsilon-greedy and off-policy), TD Learning (Q-Learning and SARSA) and Q-Learning with Function Approximation.
- Implemented Deep Q-Learning and Policy Gradient methods for Atari Games using PyTorch and OpenAI Gym.

Image Captioning with Visual Attention

IIT Kanpur

PROGRAMMING CLUB, IIT KANPUR

February 2018 - May 2018

- Compared various CNN based models for image classification and implemented them using eager execution in Tensorflow.
- Implemented the model described in Show, Attend and Tell (Xu et al.2015) using Tensorflow's estimator API and evaluated the model on MS COCO dataset.

Microsoft code.fun.do hackathon

IIT Kanpur

NATIONAL-LEVEL HACKATHON WINNER

February 2018

- Selected out of 120 students to represent IIT Kanpur at Microsoft Hyderabad center to showcase our project in their academia-industry collaboration event AXLE.
- Built an interactive interface using D3.js to display changing geopolitical relations and popularity of world leaders.
- Used Scrapy, a scraping framework to extract world news.
- Performed sentiment analysis and named entity-recognition on the extracted news to infer the effect of the concerned news on world politics.

POSITIONS OF RESPONSIBILITY

Programming Club

IIT Kanpur

SECRETARY

April 2018 - April 2019

- Helped conduct and organize Linux fest, introductory workshops and various hackathons.

Association of Computing Activities

IIT Kanpur

PROJECT MENTOR

January 2019 - April 2019

- Mentored first-year students on a semester long project on Generative Adversarial Networks.

Programming Club

IIT Kanpur

PROJECT MENTOR

May 2019 - July 2019

- Mentored first-year students on a semester long project on Probabilistic Machine Learning.

Relevant Courses

Introduction to Programming - A

Probability for Computer Science - A*

Calculus and Real Analysis - A*

Discrete Mathematics - A

Logic for Computer Science - A*

Topics in Probabilistic Modeling and Inference - A

Computational Cognitive Science (audited)

Linear Algebra and ODE - A

Data Structures and Algorithms - A

Bayesian Analysis - A*

Theory of Computation - A

Introduction to Machine Learning - A*

A*: grade for exceptional performance

Skills

Programming Languages

C, C++, Python, Javascript, MATLAB, Octave, Haskell

Libraries and frameworks

Tensorflow, Pytorch, Scikit-Learn, Pillow, Keras, Numpy

Web

Flask, HTML, CSS, jQuery, MySQL

Utilities

Linux shell utilities, LaTeX, Git

Extra Curricular

- Delivered a talk on Brain-Computer Interface in an event organized by Science Coffee House, IIT Kanpur.
- Winner of Blackbox - a three hour high speed hackathon based on an Esoteric language organized by Programming Club IIT Kanpur.